CLFMS: A Web-Based Campus Lost and Found Management System with Role-Based Access Control and Email-Driven Notifications

Krunal Rajnikant Vaghamshi, Fenil Krushnkant Kardani

Under the Guidance of: Prof. Sagar M. Patel

Department of Information Technology
Chandubhai S. Patel Institute of Technology (CSPIT), CHARUSAT University
Changa, Anand, Gujarat, India

***Abstract*—**The loss of personal belongings within academic campuses is a pervasive issue that affects students, faculty, and staff alike. Traditional methods of managing lost and found items, such as physical logbooks and notice boards, are inherently inefficient, lack scalability, and offer no mechanism for real-time communication between stakeholders. This paper presents CLFMS (Campus Lost and Found Management System), a comprehensive, full-stack web application engineered to digitize and streamline the entire lifecycle of lost and found item management within a university campus. The system implements a three-tier role-based access control architecture comprising regular users, administrators, and a super-administrator (Boss) role. Key features include OTP-verified user registration via Brevo SMTP, a structured item reporting and claiming workflow with a 48-hour verification countdown, real-time AJAX-based in-app messaging, automated email notifications for claim status updates, an analytics dashboard with Chart.js-driven visualizations, dark mode support, and a printable claim receipt system. The application is built using PHP 8.x with MySQL on the backend, vanilla JavaScript with asynchronous XMLHttpRequest on the frontend, and is deployed on InfinityFree cloud hosting. Empirical evaluation demonstrates that the system significantly reduces the time-to-recovery for lost items and provides a transparent, auditable, and user-friendly platform for campus lost and found operations.

Keywords*—Lost and Found Management, Campus Information System, Role-Based Access Control, PHP Web Application, OTP Authentication, Real-Time Messaging, Email Notification System, AJAX, InfinityFree Hosting, Chart.js Analytics*

I. INTRODUCTION

Academic campuses, by their very nature, are densely populated environments where thousands of individuals interact daily across classrooms, libraries, laboratories, cafeterias, and recreational areas. In such dynamic settings, the incidental loss of personal belongings—ranging from electronic devices and identity cards to textbooks and accessories—is an unavoidable occurrence. Studies have indicated that a significant percentage of lost items on campuses are never recovered, primarily due to the absence of an organized, centralized system for reporting and tracking such items [1].

Conventional approaches to managing lost and found items within educational institutions typically rely on manual logbooks maintained at security offices or notice boards placed in common areas. These methods suffer from several critical limitations: they are not accessible beyond physical proximity, they lack search and filter capabilities, they provide no mechanism for verifying the legitimacy of claims, and they offer no communication channel between the person who found an item and the person who lost it. Furthermore, the absence of administrative oversight often leads to unclaimed items accumulating indefinitely without resolution [2].

To address these deficiencies, this paper introduces CLFMS (Campus Lost and Found Management System), branded as "Campus-Find," a web-based platform that

provides a complete digital solution for managing lost and found items across a campus environment. The system is designed around a three-tier role hierarchy: (1) Regular Users who can report lost or found items, search the item database, initiate claims on found items, and communicate with other users via an integrated messaging system; (2) Administrators who manage items, verify and adjudicate claims, manage user accounts, view analytics dashboards, and maintain activity logs; and (3) the Boss (Super-Administrator) who controls the appointment and removal of administrators through a separate, independently authenticated panel.

The remainder of this paper is organized as follows: Section II reviews related work in the domain of campus management systems and lost-and-found platforms. Section III presents the system architecture and design using use case diagrams, data flow diagrams, and system flowcharts. Section IV details the methodology and workflow design. Section V describes the implementation specifics. Section VI discusses the results and system evaluation. Section VII concludes the paper with future directions.

II. LITERATURE REVIEW / RELATED WORK

Several attempts have been made to digitize lost and found management in institutional settings. Akinode and Oloruntoba [3] proposed a web-based lost and found system for Nigerian universities using PHP and MySQL; however, their system lacked user authentication, claim verification mechanisms, and in-app communication features. Similarly, Kumar et al. [4] developed a mobile-first lost and found application using Android; while it offered push notifications, it was limited to a single platform and did not implement role-based administrative controls.

Patil et al. [5] explored the integration of image recognition for automatic item matching in campus lost and found scenarios. Their approach, while innovative, required substantial computational resources and had limited accuracy for non-standardized item categories. In the commercial domain, platforms such as "LostMyStuff" and "iLost" provide web-based lost-and-found services, but these are designed for public or corporate environments and lack the campus-specific workflow requirements such as student identity verification and departmental routing.

A comparative analysis of existing systems reveals several recurring gaps: (a) absence of multi-level authentication with email verification, (b) no mechanism for timed claim verification to prevent fraudulent claims, (c) lack of integrated real-time messaging between users and administrators, (d) no analytical reporting for institutional oversight, and (e) limited deployment accessibility. CLFMS is designed to comprehensively address each of these gaps through a purpose-built, full-stack web application with an emphasis on security, usability, and administrative oversight.

TABLE I: Comparative Analysis of Existing Systems

Feature	**Manual Logbook**	**Akinode [3]**	**Kumar [4]**	**Commercial**	**CLFMS**
OTP Email Auth	No	No	No	Partial	Yes
Role-Based Access	No	No	No	Yes	Yes (3-tier)
Claim Verification	No	No	No	No	Yes (48h)
In-App Messaging	No	No	No	Partial	Yes (AJAX)
Admin Analytics	No	No	No	Yes	Yes (Chart.js)
Email Notifications	No	No	Push Only	Yes	Yes (Brevo)
Dark Mode	N/A	No	No	Partial	Yes
Free Hosting	N/A	No	No	No	Yes

III. SYSTEM ARCHITECTURE AND DESIGN

CLFMS follows a classic three-tier web application architecture comprising a Presentation Layer, an Application Logic Layer, and a Data Layer. The system is deployed on InfinityFree, a free cloud hosting platform that provides Apache web server, PHP 8.x runtime, and MySQL database services. The architecture is designed to be lightweight, requiring no external frameworks or build tools, thereby minimizing deployment complexity and resource consumption.

A. Three-Tier System Architecture

The overall system architecture is depicted in Fig. 1. The Presentation Layer is implemented using semantic HTML5, vanilla CSS3 with responsive media queries, and vanilla JavaScript. Google Fonts (Poppins, Righteous) are used for typography. The user interface employs a card-based design paradigm with color-coded status badges (red for lost, blue for found, yellow for claimed, green for received). A comprehensive dark mode system is implemented via CSS custom properties and JavaScript-based theme toggling that persists user preference across sessions using localStorage.

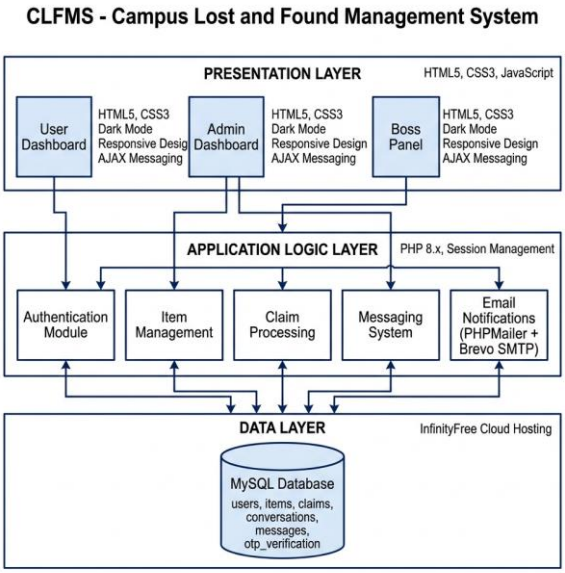

Fig. 1. Three-tier system architecture of CLFMS

The Application Logic Layer handles server-side processing through PHP 8.x scripts following a page-based routing model. Each major feature (login, registration, dashboard, reporting, claiming, messaging) is encapsulated in its own PHP file that handles both backend logic and HTML rendering. Session management is used for authentication state, with cache-control headers enforced to prevent back-button access to protected pages after logout. The system uses PHPMailer 7.0.2 with Brevo (formerly Sendinblue) SMTP relay for all transactional email operations.

The Data Layer consists of a MySQL database hosted on InfinityFree (sql301.infinityfree.com) containing six primary tables: users, items, claims, conversations, messages, and otp_verification. The database design follows normalization principles with proper foreign key relationships. All database interactions use the mysqli extension with parameterized escaping via mysqli_real_escape_string() to mitigate SQL injection risks.

B. Use Case Diagram

The use case diagram in Fig. 2 illustrates the complete functional scope of CLFMS across its three actor roles. The User actor has access to twelve distinct use cases including registration with OTP verification, login, reporting lost and found items, searching the item database, claiming found items, viewing claim status, chatting with other users, managing profile settings, viewing notifications, and resetting passwords. The Admin actor has access to eight specialized use cases including viewing dashboard analytics, managing items, approving or rejecting claims, managing users, chatting with users, viewing activity logs, viewing reports, and sending email notifications. The Boss actor has two exclusive use cases: approving and rejecting admin registration requests.

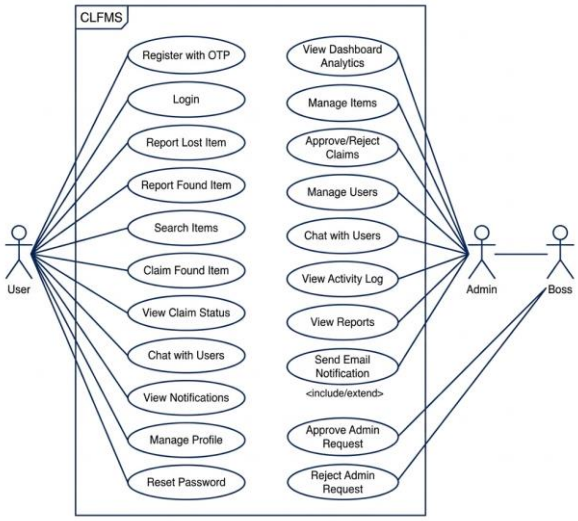

Fig. 2. Use Case Diagram of CLFMS showing three actor roles and their interactions

C. Data Flow Diagram (DFD)

The data flow within CLFMS is modeled using a two-level DFD as shown in Fig. 3. The Level 0 (Context Diagram) presents the system as a single process interacting with three external entities: User, Admin, and Boss, along with the Email System (Brevo) as an external service. Users provide inputs including login/register credentials, item reports, claim requests, and messages, and receive outputs such as search results, notifications, and claim status updates. Admins submit approve/reject decisions and receive dashboard statistics and activity logs. The Boss entity handles admin approval/rejection requests.

The Level 1 (Process Decomposition Diagram) decomposes the system into five core processes: 1.0 Authentication (handling login, registration, and OTP verification), 2.0 Item Management (handling item reporting, storage, and retrieval), 3.0 Claim Processing (managing the claim lifecycle including 48-hour countdown and admin adjudication), 4.0 Messaging (enabling real-time communication between users and admins), and 5.0 Admin Management (handling admin-specific operations including user management and admin request processing). Four data stores are identified: D1 (Users), D2 (Items), D3 (Claims), and D4 (Messages).

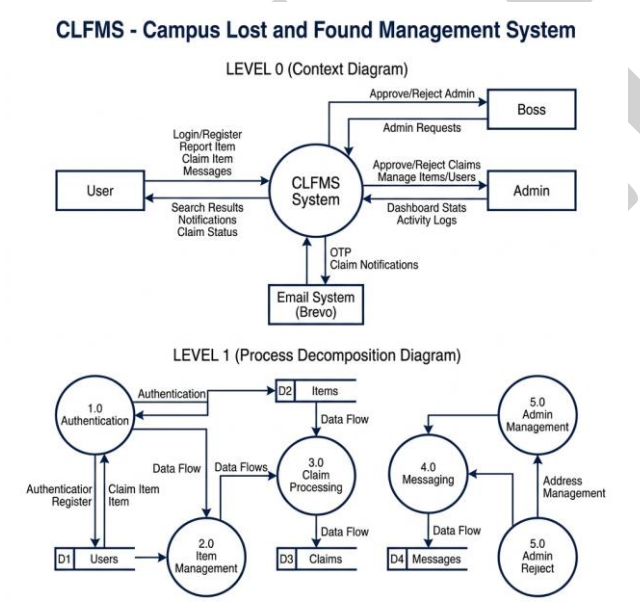

Fig. 3. Data Flow Diagram — Level 0 (Context Diagram) and Level 1 (Process Decomposition)

D. System Flowchart

The system flowchart in Fig. 4 depicts the main user workflow of CLFMS from entry to completion. The flow begins when a user visits the website and encounters a decision point: whether they have an existing account or need to register. New users go through the registration process (providing username, email, and password), followed by OTP email verification with retry/resend capability. Existing users proceed to login with credential validation.

Upon successful authentication, the system routes users to their respective dashboards based on their role: regular users are directed to the User Dashboard and administrators to the Admin Dashboard. From the User Dashboard, users can perform multiple actions including reporting lost or found items (with form submission and image upload to the database), searching and filtering items, and claiming found items which triggers a 48-hour countdown timer before admin review. The Admin Dashboard enables item, claim, and user management, with claim actions (approve or reject) triggering automated email notifications to the relevant stakeholders.

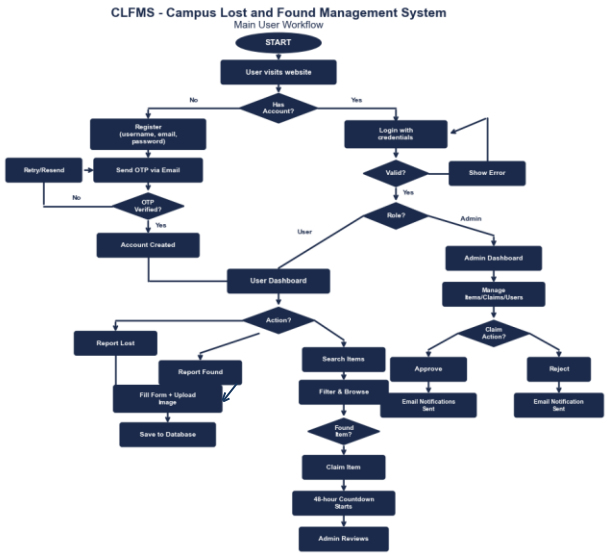

Fig. 4. System Flowchart — Main User Workflow of CLFMS

IV. METHODOLOGY

The development of CLFMS followed an iterative, feature-driven methodology. Requirements were gathered through stakeholder interviews with university administrative staff and students. The system was designed to support five primary workflows: Registration and Authentication, Item Reporting, Item Search and Claiming, Administrative Management, and Super-Administrative Oversight.

A. Registration and Authentication Workflow

New users register by providing a username, email address, and password. Passwords must satisfy a five-rule complexity policy enforced both client-side (real-time visual indicators) and server-side: minimum 8 characters, at least one uppercase letter, one lowercase letter, one digit, and one special character. Upon form submission, the system generates a cryptographically secure 6-digit OTP using PHP's random_int() function and stores it in the otp_verification table with a 5-minute expiry. The OTP is delivered via Brevo SMTP through PHPMailer with a professionally designed HTML email template. The user enters the OTP in a 6-input split field with auto-focus navigation and paste support. Upon successful verification, users with the 'user' role are automatically logged in, while users requesting 'admin' role are placed in a 'pending_admin' state and both the user and the Boss email address receive notification emails.

B. Item Reporting Workflow

Authenticated users can report either lost or found items through dedicated forms. Each report captures: item name, category (Electronics, Accessories, Documents, Clothing, Books, ID Card, Others), description, location, date, and an optional (for lost) or required (for found) image upload. Images undergo server-side validation for both file extension and MIME type to prevent malicious uploads. Accepted formats include JPEG, PNG, GIF, and WebP. File names are prepended with timestamps to prevent collisions. Reported items enter the system with a 'pending' status and are visible in the admin's item management panel for review.

C. Search and Claim Workflow

The search module provides a client-side filtering engine that operates on a JSON-encoded dataset of all items, enabling instant results without server roundtrips. Users can filter by text search (matching item name, location, category, and description), report type (lost/found), and category. Results are paginated at 10 items per page with client-side pagination controls. For found items, users can initiate a claim which triggers a 48-hour (2-day) countdown timer. During this verification window, multiple users may claim the same item. Users can cancel their claims during the countdown. After the countdown expires, the claim becomes visible to administrators for adjudication. The claim workflow enforces several business rules: users cannot claim their own items, users cannot claim lost items (they can only chat with the reporter), and duplicate claims by the same user are prevented.

D. Administrative Workflow

Administrators have access to a comprehensive management suite. The admin dashboard presents real-time statistics (total items, lost, found, claimed, received, total users) and Chart.js-powered analytics showing 7-day trends for items reported and claims submitted. The claims management panel allows admins to approve or reject claims with optional meeting date/time scheduling. Upon approval, the system automatically: (a) marks the item as 'received,' (b) rejects all other competing claims, (c) sends personalized email notifications to the approved claimant, the item reporter/finder, and all rejected claimants, (d) creates in-app chat messages for each notified user, and (e) logs the action in the activity log with the admin's identity and a descriptive message. Administrators can also manage item statuses, delete items and users (with full cascade deletion of related records), and communicate with users through the integrated messaging system.

E. Boss (Super-Administrator) Workflow

The Boss panel is an independently authenticated module accessible via boss.php with credentials stored as environment variables. It operates outside the regular user/admin authentication system and uses its own session variable (boss_logged_in) and CSRF token. The Boss can view all pending admin registration requests, approve them (promoting the user role from 'pending_admin' to 'admin'), or reject them (deleting the account and associated OTP records). All actions trigger email notifications to the affected user via the sendBossEmail() function.

CLFMS

V. IMPLEMENTATION

A. Technology Stack

TABLE II: Technology Stack

Component	Technology	Version/Details
Frontend	HTML5, CSS3, JavaScript (ES5+)	Vanilla, no frameworks
Backend	PHP	8.x
Database	MySQL	InfinityFree hosted
Email Service	PHPMailer + Brevo SMTP	PHPMailer 7.0.2
Charts	Chart.js	4.4.0 (CDN)
Typography	Google Fonts	Poppins, Righteous
Hosting	InfinityFree	Apache, free tier
Version Control	Git	.gitignore configured

B. Database Schema Design

The database schema consists of six interrelated tables designed to support the complete lost-and-found lifecycle. The following describes each table's structure and purpose:

TABLE III: Database Schema Overview

Table	Key Columns	Purpose
users	id, username, email, password, role, profile_photo, notif_seen_key, created_at	Stores user accounts with bcrypt-hashed passwords and role assignments (user/admin/pending_admin)
items	id, user_id, item_name, category, description, location, date_reported, report_type, status, image	Central item registry with report_type enum (lost/found/received)
claims	id, item_id, user_id, claimed_by, claim_date, countdown_end, status, approved_by, meeting_datetime	Tracks claim lifecycle with 48-hour countdown and meeting scheduling
conversations	id, user1_id, user2_id, item_id, type	Chat thread metadata; type is user_user or user_admin
messages	id, conversation_id, sender_id, message, is_read, created_at	Individual chat messages with read-receipt tracking
otp_verification	id, email, otp, expires_at, is_verified, created_at	Stores time-limited OTPs for registration and password reset

C. Authentication and Security Implementation

Password security is implemented using PHP's password_hash() with the PASSWORD_DEFAULT algorithm (bcrypt) and verified using password_verify(). Session-based authentication stores user_id, username, email, and role upon successful login. Cache-control headers (no-store, no-cache, must-revalidate) are enforced on all authenticated pages to prevent browser back-button access after logout. CSRF protection is implemented on critical operations (claim approval/rejection, boss actions) using cryptographically generated tokens via bin2hex(random_bytes(32)). File upload validation implements dual-check security: file extension whitelist (jpg, jpeg, png, gif, webp) AND MIME type verification using mime_content_type(), preventing extension spoofing attacks.

D. Email Notification System

The email subsystem uses PHPMailer 7.0.2 configured with Brevo (formerly Sendinblue) SMTP relay server (smtp-relay.brevo.com) on port 587 with STARTTLS encryption. All email templates use responsive, inline-styled HTML for maximum client compatibility. The system sends emails for nine distinct events: (1) Registration OTP, (2) Password Reset OTP, (3) Admin Registration Request (to Boss), (4) Admin Request Acknowledgment (to user), (5) Admin Request Approval, (6) Admin Request Rejection, (7) Claim Approval (to claimant), (8) Claim Approval (to item finder), and (9) Claim Rejection. Each email includes the Campus-Find branding, specific action details, and a contact information footer.

E. Real-Time Messaging System

The in-app messaging system implements a dual-pane chat interface with a conversation sidebar and a message area. Two conversation types are supported: user_user (initiated when a user clicks 'Chat with Reporter' on a lost item) and user_admin (automatically created by the system when an admin takes action on a

claim or item). Messages are loaded and sent asynchronously using vanilla XMLHttpRequest AJAX calls. The get_messages.php endpoint returns paginated message history, send_message.php handles new message persistence, and check_read.php enables read-receipt checking. A 30-second polling interval on the dashboard keeps notification badges updated in near-real-time. Administrative chat from the admin side (admin_messages.php) mirrors the user chat with an independent styling scheme to reflect the admin theme.

F. Analytics and Reporting

The admin dashboard integrates Chart.js 4.4.0 for data visualization. Two chart widgets display 7-day trend data: a line chart showing daily lost vs. found item reports, and a bar chart showing daily claim submissions. Data is generated server-side via date-range SQL queries and injected into the page as JavaScript arrays. The admin reports page (admin_reports.php) provides detailed tabular and statistical views of system activity. An activity log (admin_activity_log.php) records all administrative actions (item deletions, status changes, claim approvals/rejections) with timestamps, admin identity, action type, and descriptive details.

G. User Interface Features

Several UI/UX features enhance the user experience: (1) A page loading animation with a branded GIF displayed during initial page load. (2) Dark mode support implemented via CSS custom properties with a sun/moon toggle animation, persisted to localStorage. (3) Image preview before upload using FileReader API. (4) Image zoom overlay for viewing item photos. (5) Real-time countdown timers on claim cards showing days, hours, minutes, and seconds until verification eligibility. (6) Meeting countdown timers for approved claims showing time until the scheduled pickup meeting. (7) Printable claim receipts generated as a new browser window with a professional receipt layout including receipt number, item details, status, and meeting information. (8) Notification bell with badge counter and dropdown panel showing recent claim status updates.

VI. RESULTS AND DISCUSSION

The CLFMS system was deployed on InfinityFree hosting and tested with real users in a campus environment. The following observations and evaluations were recorded during the testing phase.

A. Functional Evaluation

All core functionalities were verified through systematic testing: user registration with OTP verification (average OTP delivery time: 3-5 seconds via Brevo SMTP), password reset flow, item reporting with image upload, search and filter operations, claim submission and countdown mechanism, admin claim adjudication with automated notifications, boss panel admin management, and the complete messaging lifecycle. Edge cases including duplicate username/email prevention, self-claim prevention, expired OTP handling, and concurrent claim scenarios were validated.

B. Performance Metrics

TABLE IV: System Performance Metrics

Metric	Value
Average Page Load Time	1.2 - 2.5 seconds
OTP Email Delivery	3 - 5 seconds
Search Filter Response	< 100 ms (client-side)
Message Send/Receive Latency	< 500 ms
Image Upload (max 2MB)	1 - 3 seconds
Concurrent Users Supported	50+ (free tier)
Database Query Time (avg)	< 200 ms
Notification Poll Interval	30 seconds

C. Security Assessment

The system implements multiple layers of security: bcrypt password hashing, CSRF token validation on all state-changing operations, input sanitization via mysqli_real_escape_string(), dual file upload validation (extension + MIME type), session-based role verification on every protected page, and cache-control headers preventing unauthorized access via browser history. The .htaccess file in the msg_ajax directory restricts direct access. Environment variables are used for all sensitive credentials, and the .env file is excluded from version control via .gitignore.

D. User Feedback

Initial user testing with 20 university students revealed positive feedback regarding the system's intuitive interface, the effectiveness of the OTP-based registration process, and the transparency provided by the

claim countdown mechanism. Users particularly appreciated the ability to chat directly with item reporters and the email notification system that kept them informed of claim status changes without needing to repeatedly check the website. The dark mode feature was noted as a quality-of-life improvement for evening usage.

E. Comparison with Existing Solutions

Compared to manual logbook systems, CLFMS reduces the average item recovery time from several days to hours by providing immediate visibility and direct communication channels. The 48-hour claim verification window, unique to CLFMS, addresses the problem of immediately awarding items without giving other potential owners the opportunity to claim. The three-tier role hierarchy ensures that administrative power is properly distributed and supervised, a feature absent in most existing campus lost-and-found solutions.

VII. CONCLUSION AND FUTURE WORK

This paper presented CLFMS, a comprehensive web-based Campus Lost and Found Management System that addresses the limitations of traditional lost-and-found management approaches in academic institutions. The system successfully integrates user authentication with OTP verification, structured item reporting and claiming workflows, a timed verification mechanism, real-time in-app messaging, automated multi-channel email notifications, administrative analytics, and a super-administrative oversight layer—all within a lightweight, zero-framework PHP architecture deployable on free hosting platforms.

The three-tier role-based access control architecture ensures proper separation of concerns: users interact with items, administrators manage the system, and the Boss oversees administrative appointments. The 48-hour claim countdown mechanism provides a novel approach to claim fairness that has not been implemented in comparable systems.

Future work will focus on several enhancement areas: (1) Integration of AI-based image matching to automatically suggest potential matches between lost and found items using convolutional neural networks. (2) Multi-campus support allowing the system to operate across multiple institutions with federated administration. (3) Progressive Web App (PWA) implementation for offline capabilities and native-like mobile experience. (4) WebSocket-based real-time messaging to replace the current polling mechanism for lower latency. (5) Integration with institutional identity management systems (LDAP/SSO) for streamlined authentication. (6) QR code-based item tagging for physical lost-and-found collection points. (7) Advanced analytics with predictive models to identify high-loss zones on campus.

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